

Machine learning

Machine learning is a branch of artificial intelligence that deals with the self-learning of computers. This means that machines effectively learn to grow and change without any need for programming.

And automation is the new emerging technology through which no more human assistance is required during any process. In other words, this is programmed control. This process uses different control frameworks for different systems or equipment such as electronic hardware, production lines, boilers, phone systems, boats, airplanes and vehicles, so that there is insignificant or less human mediation. A few procedures have already been totally computerised.

The automation, which we know today, began with simple machines, like water wheels, in the 11th century. Gradually, humans started developing on these simple machines and innovating over time.

Open source artificial intelligence tools

Now let's look at a few open source AI tools that are helping researchers currently working on their projects.

Apache Mahout: This is an open source machine learning tool under the Apache licence. It has been designed to simplify working on common math problems involving statistics and linear algebra. This ML tool has been designed by the Apache Software Foundation and is built on a platform called Mahout Hadoop. Data analysts use this open source machine learning tool to carry out a critical and detailed analysis of Big Data. An understanding of quantitative data helps to extract useful information and pinpoint the trends. Mahout's algorithms help users in clustering and grouping Big Data by first saving the data in the Hadoop distributed file system. It has a cross-platform operating system, and it gives users an environment with R-like syntax.

Link: <https://mahout.apache.org/>

Distributed Machine Learning Toolkit (DMTK): This is an open source ML tool that simplifies various tasks on Big Data. DMTK was released by Microsoft, with new and advanced algorithms getting added to this toolkit regularly. This toolkit lets researchers experiment with algorithms by modifying them and tweaking them according to their needs. Previously, researchers needed many computational resources to work on Big Data, since it has been proved that bigger data models lead to better results, but this was quite challenging. The DMTK enables innovation in both the system's algorithm and in ML algorithms.

Link: <http://www.dmtk.io/>

Open Neural Networks (OpenNN): This is an open source library written in the C++ language. It has been

designed for deep learning and for the implementation of neural networks. This library has an easy-to-understand and deep architecture, and is used in advanced ML research. This open source library is used in the logistics and marketing verticals. OpenNN also allows high performance computing as it has a higher processing speed.

Link: <http://www.opennn.net/>

OpenCyc: This is an open source general knowledge database that makes text understanding possible. Cycorp launched OpenCyc to make sure that users have unrestricted access to this knowledge base and that they can use OpenCyc in different applications. The vast knowledge base of OpenCyc contains a diverse range of concepts, facts, taxonomies, assertions and rules. It is available in two forms, i.e., ontology as well as the semantic Web endpoints. The latter can also be referred to as the permanent URIs, which means that both the RDF representations and human-readable forms are returned to the users.

Cycorp enables apps to sift through a large database and process the relevant information to come up with an accurate interpretation. It helps in the differentiation of relative words and synonyms of a particular key search word and ensures the app performs like a person, exhibiting human cognitive abilities and discernment. OpenCyc has assisted AI researchers in their projects so they can now use its ontologies to ensure machine learning (ML) that can be interrupted by humans to prevent machines from causing harm. Cyc has revolutionised ML by enabling the apps to learn the complicated reasoning and logic of the human mind.

Link: <http://www.cyc.com/opencyc/>

H2O: This open source ML software tool, designed by H2O.ai, is used by developers and AI researchers. It is written in the R, Python and Java programming languages, and is being used for predictive data analytics by AI researchers and developers in whichever language platform they're familiar with. It can also be used to analyse data sets in the cloud and in Apache Hadoop file systems. It supports different operating systems — Linux, MacOS and Microsoft Windows.

This open source deep learning platform helps in decision making after the deep analysis of data so that the user can draw useful insights. Having two open source versions, this tool has found widespread application in predictive modelling, healthcare and fraud analysis.

Link: <https://www.h2o.ai/>

TensorFlow: This open source ML library was launched by Google Brain. It lets you write libraries for dataflow programming. It operates in two programming languages — C++